



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2013

Year 11 Mathematics
Yearly

Mathematics

General Instruction

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen.
Pencil may be used for diagrams.
- Board approved calculators maybe used.
- Start each **NEW** question in a separate answer booklet.
- Marks may **NOT** be awarded for messy or badly arranged work.
- All necessary working should be shown in every question.
- Answer in simplest exact form unless otherwise instructed.

Total Marks – 70

- **Attempt Questions 1 – 6**

Examiner: *R. Elliott*

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

START A NEW ANSWER BOOKLET

QUESTION 1 (10 marks)

- (a) Simplify $4x^3y^{-2} \div 2xy^{-3}$. 1
- (b) What is the gradient of $2x + 3y = 5$? 1
- (c) If $(2, a)$ is on the line $2x + 3y = 5$ find the value of a . 1
- (d)
- (i) Find where the curve $y = x^2 - x - 6$ cuts the x -axis. 2
- (ii) What is the minimum value of the same function? 2
- (e) Simplify $3^x \times 5^x$. 1
- (f) Solve $\frac{x}{2} - \frac{x-1}{3} = 5$. 2

End of Question 1

START A NEW ANSWER BOOKLET

QUESTION 2 (11 marks)

- (a) On separate graphs sketch the following. 4
- (i) $y = 2 - x^2$
 - (ii) $xy - 3 = 0$
 - (iii) $y = |x + 1|$
 - (iv) $y = 3^{-x}$
- (b) For the parabola $x^2 = 4y - 8$ write down the coordinates of 4
- (i) the focus
 - (ii) the vertex
 - (iii) the equation of the directrix
 - (iv) the equation of the line through the focus parallel to the directrix.
- (c) Solve the following quadratic equations 3
- (i) $2x^2 - 5x - 3 = 0$
 - (ii) $2x^2 - 4x - 1 = 0$ (leave answer as a surd)

End of Question 2

START A NEW ANSWER BOOKLET

QUESTION 3 (12 marks)

- (a) Given α and β are the roots of $2x^2 + 3x - 4 = 0$ find the values of 4
- (i) $\alpha + \beta$
 - (ii) $\alpha\beta$
 - (iii) $\alpha^{-1} + \beta^{-1}$
 - (iv) $\alpha^3 + \beta^3$
- (b) A triangle has sides of length 7, 5 and 4 cm.
- (i) Find the smallest angle. 2
 - (ii) Find its area. 2
- (c) Solve each of the following equation and graph the solutions on a number line.
- (i) $|x + 2| = 7$ 2
 - (ii) $3x + 15 \leq 0$ 2

End of Question 3

START A NEW ANSWER BOOKLET

QUESTION 4 (13 marks)

(a) Find derivatives of the following 5

(i) $x^2 - \frac{2}{x} + 5$

(ii) $x(x^2 - 3)^3$

(iii) $\frac{x}{\sqrt{x}-1}$

(b) Find the equation of the tangent drawn to the curve $y = \sqrt{x} + 1$ at the point where $x = 4$. 2

(c) Determine whether any of the following are odd, even or neither functions, justifying your answer. 6

(i) $f(x) = x\sqrt{x^2 - 1}$

(ii) $f(x) = \frac{4}{x^2 - x}$

(iii) $f(x)(x + 2) = 1$

End of Question 4

START A NEW ANSWER BOOKLET

QUESTION 5 (11 marks)

- (a) Graph on the number plane the region that is solved simultaneously by 3

$$y - 1 > |x| \quad \text{and} \quad y \leq x^2 + 2$$

- (b) If $\log a = 1.47$ and $\log b = 0.86$ find correct to 2 decimal places 2

(i) $\log\left(\frac{a}{b}\right)$

(ii) $\log(\sqrt{ab})$

- (c) Solve for x 3

$$\log_3\left(\frac{x^2}{x-2}\right) = 2$$

- (d) Factorise fully 3

(i) $2x^2 - 4x - 30$

(ii) $x^3 - y^3 - x + y$

End of Question 5

START A NEW ANSWER BOOKLET

QUESTION 6 (13 marks)

- (a)
- (i) Find the 37^{th} term of the series 2
- $$19 + 15 + 11 + \dots$$
- (ii) Find the sum of these 37 terms. 2
- (b) Find the 19^{th} term of the series $96 - 48 + 24 - \dots$ (leave answer in simplified index form) 2
- (c) Express $0.\dot{1}\dot{7}$ as a geometric series and hence write it as a fraction. 3
- (d) A boy starts work in a Hamburger shop and his father tells him to save \$10 out of his first week's salary. He then tells him to save an additional 50 cents each week from then on. In other words he saves \$10.50 in the second and \$11 in the third and so on. How much does he save in a year. 4

End of Question 6
End of Exam

Y11 Mathematics Yearly - Solutions (2013)

Question 1 (10 marks)

$$(a) \frac{4x^3}{y^2} \div \frac{2x}{y^3} = \frac{4x^3}{y^2} \times \frac{y^3}{2x} \\ = 2x^2y \quad (1)$$

$$(b) 2x + 3y = 5 \\ \Rightarrow (y = mx + b) \\ 3y = 5 - 2x \\ y = \frac{5}{3} - \frac{2x}{3} \\ \therefore m = -\frac{2}{3} \quad (1)$$

$$(c) 2x + 3y = 5, (2, a) \\ 2(2) + 3a = 5 \\ 4 + 3a = 5 \\ 3a = 1 \\ a = \frac{1}{3} \quad (1)$$

$$(d) (i) x\text{-intercepts } (y=0) \\ \therefore x^2 - x - 6 = 0 \\ (x-3)(x+2) = 0 \\ x-3=0 \text{ or } x+2=0 \\ x=3 \quad x=-2 \\ \therefore (3,0) \text{ and } (-2,0) \quad (1)$$

(ii) $a > 0 \therefore$ minimum value at vertex.

$$x = \frac{-b}{2a} \Rightarrow x = \frac{-(-1)}{2(1)} \\ x = \frac{1}{2} \quad (1)$$

$$y = \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right) - 6 \\ = -6\frac{1}{4}$$

$$\therefore \left(\frac{1}{2}, -6\frac{1}{4}\right) \quad (1)$$

$$(e) 3^x \times 5^x = 15^x \quad (1)$$

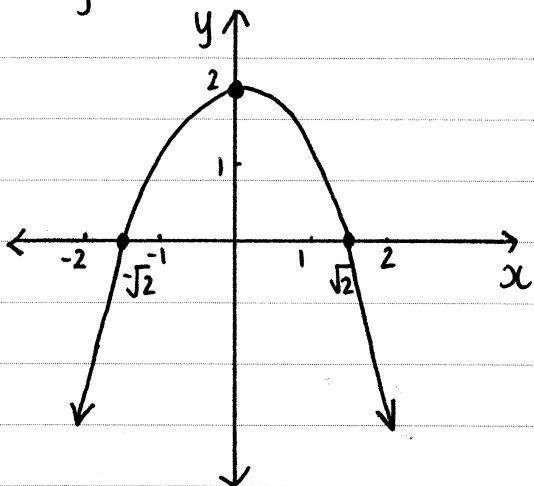
$$(f) \frac{x}{2} - \frac{x-1}{3} = 5 \\ 3x - 2(x-1) = 30 \quad (1) \\ 3x - 2x + 2 = 30 \\ x + 2 = 30 \\ x = 28 \quad (1)$$

Question 2

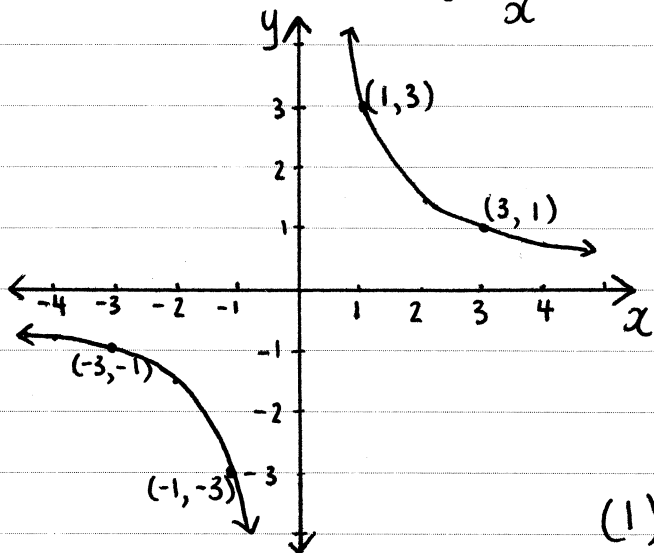
(11 marks)

(a)

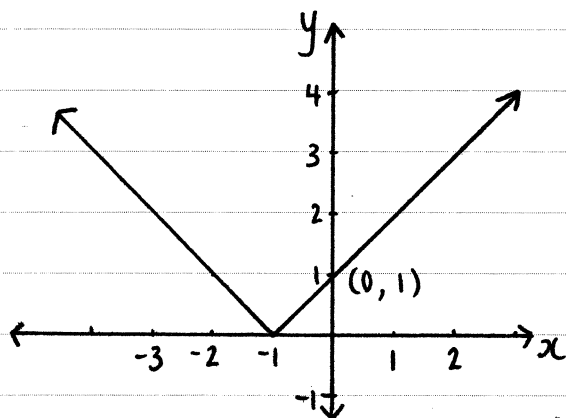
(i) $y = 2 - x^2$



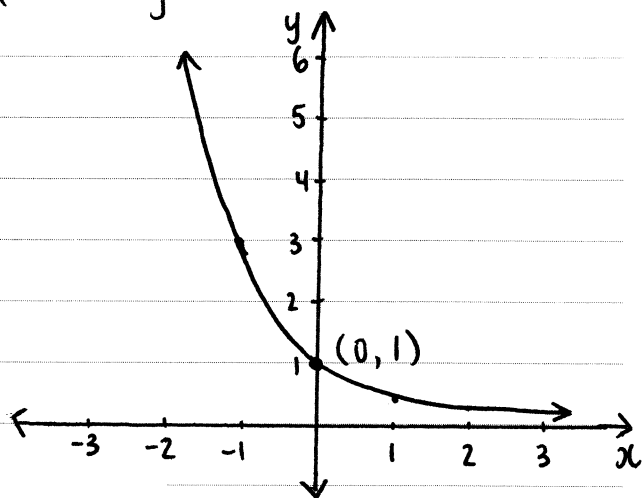
(ii) $xy - 3 = 0 \Rightarrow y = \frac{3}{x} \quad x \neq 0$



(iii) $y = |x + 1|$



(iv) $y = 3^{-x}$



(1) (b) $x^2 = 4y - 8$

$x^2 = 4(y - 2)$

$x^2 = 4(1)(y - 2)$

(i) focus $\Rightarrow$

focal length $a = 1$

$\therefore$ Focus at $(0, 2 + 1)$

Focus $(0, 3)$

(1)

(ii) Vertex $(0, 2)$

(1)

(iii) directrix with equation

$y = 2 - 1$

$y = 1$

(1)

(iv) $y = 3$

(1)

(c)

$$(i) 2x^2 - 5x - 3 = 0$$

$$\begin{matrix} P: -6 \\ S: -5 \end{matrix} \left. \vphantom{\begin{matrix} P: -6 \\ S: -5 \end{matrix}} \right\} -b, +1$$

$$(2x - 6)(2x + 1) = 0$$

2

$$\cancel{x}(x - 3)(2x + 1) = 0$$

$\cancel{x}$

$$x - 3 = 0 \text{ or } 2x + 1 = 0$$

$$\boxed{x = 3}$$

$$2x = -1$$

$$\boxed{x = -\frac{1}{2}}$$

(1)

$$(ii) 2x^2 - 4x - 1 = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-(-4) \pm \sqrt{(-4)^2 - 4(2)(-1)}}{2(2)}$$

$$= \frac{4 \pm \sqrt{24}}{4}$$

$$= \frac{4 \pm 2\sqrt{6}}{4} \quad (1)$$

$$= \cancel{x} \frac{2 \pm \sqrt{6}}{2}$$

$\cancel{x} \frac{2}{2}$

$$\therefore \boxed{x = \frac{2 + \sqrt{6}}{2}} \text{ or } \boxed{x = \frac{2 - \sqrt{6}}{2}} \quad (1)$$

Question 3

(12 marks)

(a) $2x^2 + 3x - 4 = 0$

(i) $\alpha + \beta = -\frac{b}{a}$

$$= -\frac{3}{2} \quad (1)$$

(ii) $\alpha\beta = \frac{c}{a}$

$$= -\frac{4}{2}$$

$$= -2 \quad (1)$$

(iii) $\alpha^{-1} + \beta^{-1} = \frac{1}{\alpha} + \frac{1}{\beta}$

$$= \frac{\beta + \alpha}{\alpha\beta}$$

$$= -\frac{3}{2}$$

$$= -2$$
$$= \frac{3}{4} \quad (1)$$

(iv) $\alpha^3 + \beta^3$

$$= (\alpha + \beta)(\alpha^2 - \alpha\beta + \beta^2)$$

$$= (\alpha + \beta)(\alpha^2 + \beta^2 - \alpha\beta)$$

↓

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

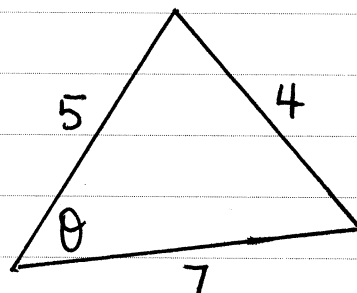
$$\therefore = (\alpha + \beta)[(\alpha + \beta)^2 - 2\alpha\beta - \alpha\beta]$$
$$= (\alpha + \beta)[(\alpha + \beta)^2 - 3\alpha\beta]$$

$$= \left(-\frac{3}{2}\right)\left[\left(-\frac{3}{2}\right)^2 - 3(-2)\right]$$

$$= \left(-\frac{3}{2}\right)\left(\frac{33}{4}\right)$$

$$= -\frac{99}{8} \quad \text{or} \quad -12.375 \quad (1)$$

(b)



(i) smallest angle θ

$$\cos \theta = \frac{7^2 + 5^2 - 4^2}{2(7)(5)} \quad (1)$$

$$\theta = \cos^{-1}\left(\frac{7^2 + 5^2 - 4^2}{2(7)(5)}\right)$$
$$= 34^\circ 2' 52''$$

$$\theta = 34^\circ \quad (1)$$

(ii) Area:

$$A = \frac{1}{2} \times 5 \times 7 \times \sin 34^\circ \quad (1)$$

$$= 9.78586 \text{ cm}^2$$

$$A = 9.8 \text{ cm}^2 \quad (1)$$

(c)

$$(i) |x + 2| = 7$$

/ \

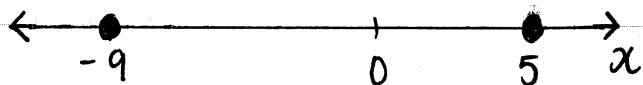
$$x + 2 = 7$$

$$x = 5$$

$$x + 2 = -7$$

$$x = -9$$

(1)



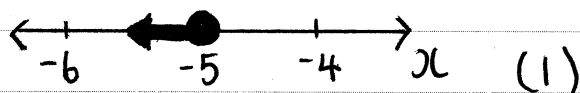
(1)

$$(ii) 3x + 15 \leq 0$$

$$3x \leq -15$$

$$x \leq -5$$

(1)



(1)

13

Solutions to 4R11 2U Maths Yearly (Sept) 2013

④ (a) (i) $\frac{d}{dx} (x^2 - 2x^{-1} + 5)$

$$= 2x + 2x^{-2} = 2x + \frac{2}{x^2} \text{ (1) //$$

(ii) $\frac{d}{dx} (x(x^2-3)^3)$

$$= x \times 3(x^2-3)^2 \times 2x + (x^2-3)^3 \times 1$$

$$= 6x^2(x^2-3)^2 + (x^2-3)^3$$

$$= (x^2-3)^2 [6x^2 + (x^2-3)']$$

$$= (x^2-3)^2 [7x^2-3] \text{ (2) //$$

(iii) $\frac{d}{dx} \left(\frac{x}{x^{\frac{1}{2}}-1} \right) \frac{u}{v}$

$$= \frac{(\sqrt{x}-1) \times 1 - x \times \frac{1}{2}x^{-\frac{1}{2}}}{(\sqrt{x}-1)^2}$$

$$= \frac{\frac{\sqrt{x}-1}{1} - \frac{x}{2\sqrt{x}}}{(\sqrt{x}-1)^2} = \frac{\frac{2x-2\sqrt{x}-x}{2\sqrt{x}}}{(\sqrt{x}-1)^2}$$

$$\frac{\sqrt{x}-2}{2(\sqrt{x}-1)^2} \text{ (2) //$$

$$= \frac{(x-2\sqrt{x})}{2\sqrt{x}(\sqrt{x}-1)^2} \times \frac{\sqrt{x}}{\sqrt{x}}$$

$$\leftarrow = \frac{x\sqrt{x}-2x}{2x(\sqrt{x}-1)^2}$$

④ (b) $y = \sqrt{x} + 1$

When $x=4$, $y = \sqrt{4} + 1 = 3$ (4, 3)

$$y = x^{\frac{1}{2}} + 1$$

$$y' = \frac{1}{2}x^{-\frac{1}{2}} = \frac{1}{2\sqrt{x}}$$

when $x=4$, $m = \frac{1}{2 \times \sqrt{4}} = \frac{1}{4}$

using $(y - y_1) = m(x - x_1)$

$$(y - 3) = \frac{1}{4}(x - 4)$$

$$y - 3 = \frac{x}{4} - 1$$

$$y = \frac{x}{4} + 2 \text{ (2) or}$$

$$4y = x + 8$$

$$x - 4y + 8 = 0$$

⑤ (i) $f(x) = x \cdot \sqrt{x^2 - 1}$

$$f(-x) = (-x) \sqrt{(-x)^2 - 1} = -x \sqrt{x^2 - 1}$$

not even

now for odd, $f(x) = -f(-x)$

$$= -[-x \sqrt{x^2 - 1}]$$

is odd (2) $= x \cdot \sqrt{x^2 - 1}$

④ (c) (i) $f(x) = \frac{4}{x^2 - x}$

now $f(-x) = \frac{4}{(-x)^2 - (-x)} = \frac{4}{x^2 + x}$

now for odd, ^{not even} $f(x) = -f(-x)$

$= \frac{-4}{x^2 + x}$ not odd

So neither // (2)

(ii) $f(x) \cdot (x+2) = 1$

$f(x) = \frac{1}{x+2}$

now $f(-x) = \frac{1}{(-x)+2} = \frac{1}{-x+2}$ not even

now for odd. $f(x) = -f(-x)$

$= \frac{-1}{-x+2}$

$= \frac{-1}{-(x-2)} = \frac{1}{x-2}$

So neither // ^{not odd} (2)

11

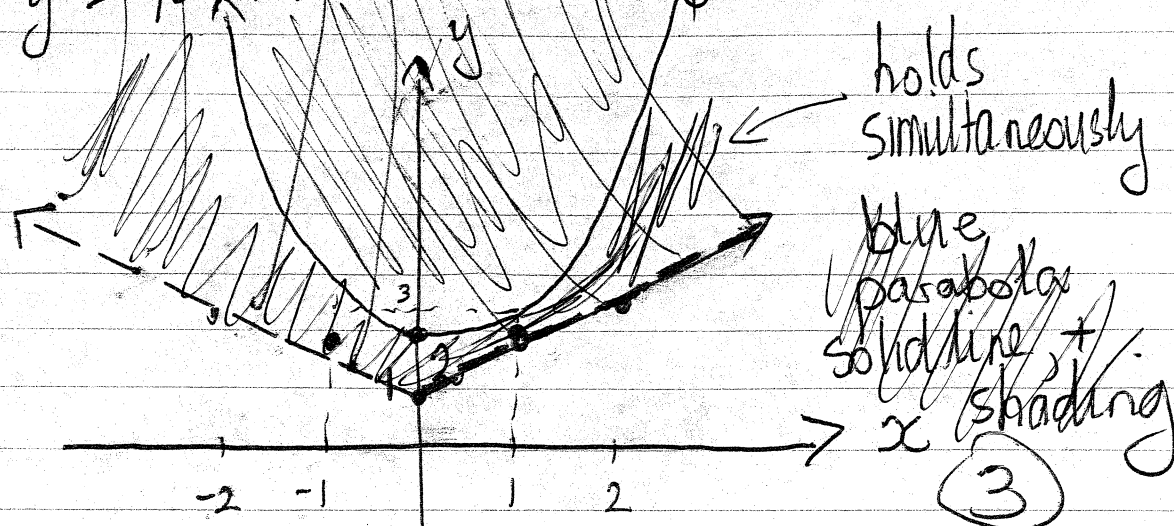
5

(a)

$$y > |x|$$

$$y > |x| + 1$$

$$\text{and } y \leq x^2 + 2$$



For $y \leq x^2 + 2$

test $(0,0)$ $0 \leq 0 + 2$ ✓

now $y > |x| + 1$

First graph $y = |x| + 1$

test $(0,0)$ $0 > |0| + 1$ NO

(b) $\log a = 1.47$, $\log b = 0.86$.

(i) $\log\left(\frac{a}{b}\right) = \log a - \log b = 1.47 - 0.86 = 0.61$

(ii) $\log(\sqrt{ab}) = \frac{1}{2} \log(ab) = \frac{1}{2} [\log a + \log b]$ ①
 $= \frac{1}{2} [1.47 + 0.86]$
 $= 1.165$ ①

$$(5) \quad (c) \quad \log_3 \left(\frac{x^2}{x-2} \right) = 2$$

$$\text{So } \frac{x^2}{x-2} = \frac{9}{1}$$

$$x^2 = 9x - 18$$

$$x^2 - 9x + 18 = 0$$

$$(x-3)(x-6) = 0$$

$$x=3, x=6. \quad (3)$$

$$(d) \quad (i) \quad 2x^2 - 4x - 30$$

$$= 2(x^2 - 2x - 15)$$

$$= 2(x-5)(x+3) \quad (1\frac{1}{2})$$

$$(ii) \quad x^3 - y^3 - x + y$$

$$x^3 - x - y^3 + y$$

$$x(x^2 - 1) - y(y^2 - 1) \text{ NO!}$$

$$\text{Back to } x^3 - y^3 - x + y$$

$$(x-y)(x^2 + xy + y^2) - 1(x-y)$$

$$= (x-y)(x^2 + xy + y^2 - 1) \quad (1\frac{1}{2})$$

13

⑥ (a) (i) $19 + 15 + 11 + \dots$

arithmetic, $a = 19$, $d = -4$ $n = 37$

$$u_n = a + (n-1)d$$

$$u_{37} = 19 + 36 \times -4 = -125 \quad (2)$$

(ii) now $S_{37} = \frac{n}{2}(a + L)$

$$= \frac{37}{2}(19 + -125) = -1961 \quad (2)$$

(b) Find u_{19} $96 - 48 + 24 - \dots$

$$a = 96, r = -\frac{1}{2}, n = 19$$

$$u_n = ar^{n-1}$$
$$= 96 \times \left(-\frac{1}{2}\right)^{n-1}$$
$$= 96 \times \left(\frac{1}{2}\right)^{n-1} \times (-1)^{n-1}$$

now $u_{19} = 96 \times \left(\frac{1}{2}\right)^{18} \times (-1)^{18}$

$$= \frac{3 \times 2^5}{2^{18}} = \frac{3}{2^{13}} \quad (2)$$

⑦ $0.\dot{1}\dot{7} = 0.17 + 0.0017 + 0.000017 + \dots$ (1)

geometric, $a = 0.17$, $r = 0.01$

$$S_{\infty} = \frac{a}{1-r} = \frac{0.17}{0.99} = \frac{17}{99} \quad (2)$$

(d) week 1 \$10 saved
week 2 \$10.50 "
week 3 \$11 "
etc.

$$a = 1000, n = 52, d = 50$$

$$S_n = \frac{n}{2} (2a + (n-1)d)$$

$$S_{52} = \frac{52}{2} (2 \times 1000 + 51 \times 50)$$

$$= \$1183 \quad (4)$$
